



Review article

Agentic AI: The age of reasoning—A review

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ARTICLE INFO

Keywords:

Agentic AI
Autonomous systems
Artificial intelligence
Large language models (LLMs)
Reasoning agents
AI taxonomy

ABSTRACT

Artificial intelligence has experienced a significant boom with the emergence of agentic AI, where autonomous agents are increasingly replacing human intervention, enabling systems to perceive, reason, and act independently to achieve specific goals. Despite its transformative potential, comprehensive information on agentic AI remains scarce in the literature. This paper provides the first comprehensive review of agentic AI, focusing on its evolution and three core aspects: patterns, types, and environments. The evolution of agentic AI is traced through five phases to the current era of multi-modal and collaborative agents, driven by advancements in reinforcement learning, neural networks, and large language models (LLMs). Five key patterns: tool use, reflection, ReAct, planning, and multi-agent collaboration (MAC) define how agentic AI systems interact and process tasks. These systems are categorized into seven categories, each tailored for specific operational styles and autonomy in decision making. The environments in which these agents operate are classified as static, dynamic, fully observable, partially observable, deterministic, stochastic, single-agent, and multi-agent, emphasizing the impact of environmental complexity on agent behavior. Agentic AI has revolutionized systems through autonomous decision making and resource optimization, yet challenges persist in aligning AI with human values, ensuring adaptability, and addressing ethical constraints. Future research focuses on multidomain agents, human–AI collaboration, and self-improving systems. This work provides researchers, practitioners, and policymakers with a structured approach to understanding and advancing the rapidly evolving landscape of agentic AI systems.

1. Introduction

The advent of agentic AI marks a transformative era in the digital landscape, redefining the role of artificial intelligence (AI) in human life and industries. Meta CEO Mark Zuckerberg has emphasized the exponential growth of AI agents, suggesting that “there could possibly be more AI agents in the world than humans”. Echoing this sentiment, NVIDIA CEO Jensen Huang envisions a paradigm shift from tools to skills, stating: “Our industry has been an industry of tools, but for the very first time, this is going to be an industry of skills. You can say the agents”. He further adds, “We are going to have agents that are excellent at particular skills”. These perspectives illuminate the unprecedented potential of agentic AI to revolutionize industries, foster innovation, and seamlessly integrate into the fabric of daily life. Agentic AI is the AI systems/programs designed to act autonomously, mimicking human-like behaviors and decision-making processes. These systems are capable of perceiving their environment, reasoning, and taking actions to achieve specific goals, exhibiting a level of agency

similar to that of human beings. Agentic AI systems are intelligent, autonomous systems that can independently make decisions, learn from data, and adapt to changing environments without relying on external support or intervention [1].

Agentic AI systems improve human action by autonomously handling complex, multitasking tasks, shifting AI from passive assistance to strategic problem solving. Unlike traditional AI confined to rigid parameters, these systems dynamically adapt to complex and dynamic evolving environments, pursuing long-term goals with minimal or no human intervention. Rather than merely executing predefined instructions, it augments human capabilities by initiating actions and making contextually informed choices, fostering collaborative innovation. As its integration accelerates, unlocking its full potential requires a rigorous exploration of its foundational principles, practical applications, and ethical-technical challenges to ensure safe and transformative deployment across industries.

Peer review under responsibility of Chongqing University.

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<https://doi.org/10.1016/j.jai.2025.08.003>

Received 9 April 2025; Received in revised form 19 August 2025; Accepted 21 August 2025

Available online 28 August 2025

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1.1. Definitions

Reasoning is logical thinking used to make a judgment. Artificial intelligence did not possess this capability in the past, but now, with the advent of agentic AI, it has begun to demonstrate reasoning abilities. The age of reasoning in AI is a transformative phase in the evolution of AI systems, in which agents (AI systems) begin to mimic aspects of human cognition. These systems are capable of generating chain-of-thought and performing well-structured, goal-directed reasoning. Moreover, these systems demonstrate the ability to adapt, plan and act with a degree of autonomy [2]. Agentic AI has been defined in various studies. According to Ref. [3], “Agentic AI” refers to systems that exhibit autonomous decision-making capabilities to achieve specific goals without constant human intervention. Another study Ref. [4] defines it as “a type of interactive system designed to interpret and respond to information from its surroundings, including images, spoken or written communication, and other contextually relevant data, while taking purposeful and contextually appropriate actions”. As noted in Ref. [5], Agentic AI systems possess rationality, enabling effective reasoning and adaptation to achieve goals. In addition, Ref. [6] describes Agentic AI as autonomous systems capable of dynamically adapting tasks, evaluation metrics, and strategies according to context, without explicit predefined instructions. Methodologically, agentic AI emphasizes key design capabilities, such as reasoning, planning, tool utilization, memory integration, retrieval-augmented generation (RAG), and instruction fine-tuning [7]. According to definitions in the literature, Agentic AI possesses several core attributes that distinguish it from classical AI systems. These include autonomy without continuous human oversight, interactive and context-aware behavior to interpret and respond to diverse inputs, rational adaptation that enables reasoning and action adjustment, dynamic strategy adjustment involving flexible task and metric reconfiguration, and advanced planning capabilities for goal-directed decision making. In addition, “the Age of Reasoning” signifies a fundamental shift from passive AI tools to active reasoning agents that can participate in knowledge work, scientific research, decision-making, and interactive problem solving, thus redefining the boundaries between human and machine intelligence [8]. Table 1 presents a comparative overview of the core capabilities that distinguish classical AI from agentic AI in key functional dimensions.

1.2. Operational framework of agentic AI

The operational framework of agentic AI systems is a cohesive process that integrates core intelligence, back-end decision making, and front-end interaction. The operational framework of agentic AI comprises four fundamental stages: Data, Analysis, Execution, and Adaptation. Fig. 1 illustrates the operational framework of an agentic AI system stages, highlighting its interaction with the back-end and front-end components to deliver adaptive and intelligent outcomes. In the Data stage, the AI Agent gathers input through various sources such as audio, visuals, and feeds information into the back-end for

initial processing. The system leverages advanced cognitive capabilities to collect and interpret data necessary to understand user intent and environmental context. During the analysis stage, the back-end infrastructure handles input validation, accesses memory, performs decision verification, and detects possible errors. This ensures a reliable and accurate determination of the optimal course of action based on stored knowledge and logic rules. The execution stage follows, where valid outputs are generated and transmitted to the front-end layer. Through API, GUI, and chatbots, the system presents responses to users, completing the task autonomously without requiring constant human oversight. Finally, in the Adaptation stage, the AI Agent engages in continuous learning by incorporating real-time feedback. This enhances the Memory module and updates future decision-making logic, enabling the system to evolve its performance and improve reliability over time.

1.3. Objective

The primary objective of this research is to provide the first comprehensive review of agentic AI, a rapidly emerging paradigm in artificial intelligence that emphasizes autonomy, decision making, adaptability, and self-improvement. Agentic AI represents a fundamental shift toward systems capable of perceiving, reasoning, planning, and executing actions independently to achieve complex objectives. Despite the increasing focus on autonomous AI agents in recent advancements, there is no existing research review paper that systematically categorizes agentic AI, analyzes its evolution, classifies and its fundamental patterns. Table 2 presents the specifications of various survey/review papers available in the literature. Most of these studies were written within the last two years and examine one dimension at a time. For example, computer vision [9], multimodal embodiment [4,10], single- vs. multi-agent architectures [11], security challenges [12], LLM agent protocols [13], scientific discovery applications [14], 6G communication systems [15], transition from generative AI to agentic AI [16], agentic AI in organizational transformation [17]. This trend indicates surveys are domain specific. The survey [6] takes agentic AI as topic, but focuses on goal-directed autonomy, RL-based architectures, adaptive control, and discusses a smaller set of applications as proof-of-concept use cases. This analysis highlights that this paper offers a distinct historical and evolutionary perspective (from the 1980s to the present day) on agentic AI, an aspect that has been largely overlooked in the existing literature. Unlike previous studies, this study provides a comprehensive analysis of agentic AI patterns, a topic that has yet to receive in-depth exploration. Although previous work typically discusses a limited number of agent types, this study delves into a broader spectrum, presenting a detailed examination of various types. This paper aims to fill this critical gap by presenting a structured review of agentic AI. This study will be foundational for research and offers a very early exploration of the agentic AI domain. This paper provides answers to the following questions:

1. What is agentic AI, and how does it redefine the role of AI in human life?

Table 1
Comparison of classical AI and agentic AI.

Capabilities	Classical AI	Agentic AI	Key difference
Autonomy	Predefined, rule-based behavior	Dynamic, self-directed behavior	Level of independence
Context-aware interaction	Static input-response behavior	Actively interprets and responds to diverse contextual inputs	Depth and quality of interaction
Adaptability	Limited, fixed scenarios	Context-sensitive, open-ended scenarios	Responsiveness to changes
Reasoning	Simple, narrow reasoning tasks	Complex reasoning and inference	Advanced strategic problem-solving
Planning	Fixed, step-by-step procedures	Flexible, goal-driven strategies	Adaptive planning toward evolving goals

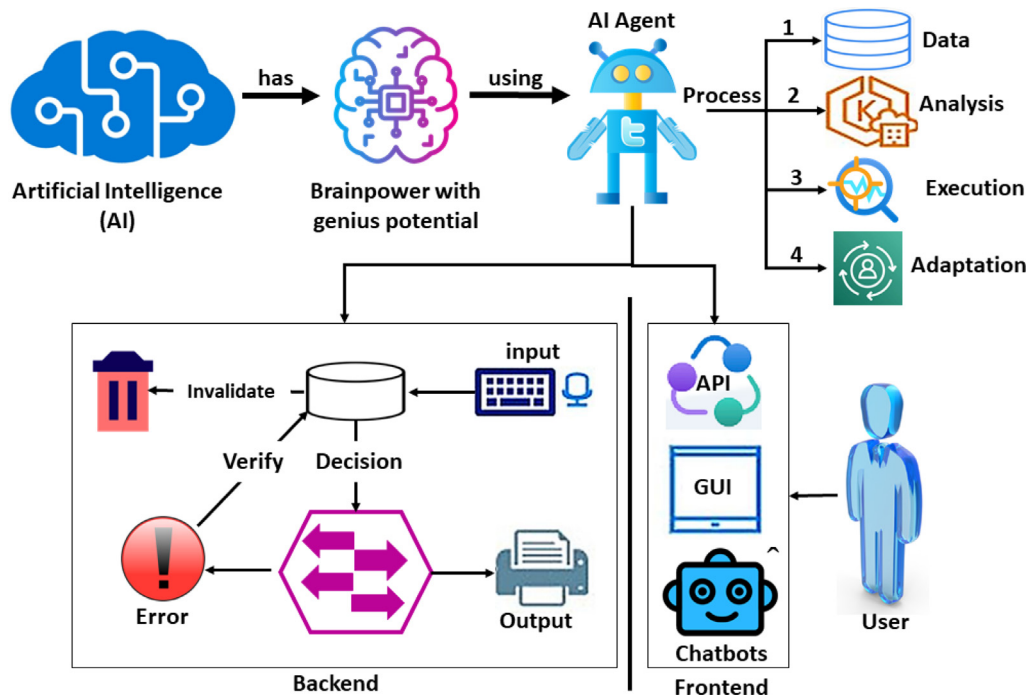


Fig. 1. The operational framework of agentic AI systems.

2. How has agentic AI evolved over time and what are its key milestones?
3. What are the key aspects of agentic AI?
4. What are the major challenges facing agentic AI and what future directions are needed to advance the field?

1.4. Paper structure

The structure of the paper is as follows:

- Section 2 provides a chronological overview of agentic AI, dividing its evolution into five distinct phases. It highlights key technological breakthroughs, influential research, and the shift from task-specific AI to general-purpose autonomous systems.
- Section 3 categorizes agentic AI systems according to their design patterns and operational styles. It provides a taxonomy to understand the diverse approaches to autonomy and decision-making, offering a framework for designing and analyzing agentic AI systems.
- Section 4 synthesizes the findings of the paper, identifies key challenges, and outlines future research directions.
- Section 5 summarizes the key contributions of the paper, emphasizing its role as the first comprehensive review and survey of agentic AI.

2. Agentic AI evolution

Agentic AI has gained momentum in recent years, but it is already reshaping the landscape of autonomous decision making, adaptive learning, and human-AI collaboration across various domains. The evolution of agentic AI has been driven by advancements in machine learning, reinforcement learning, multi-agent systems, and cognitive architectures. This paper presents a chronological overview of the field, highlighting key milestones, influential research papers, and technological breakthroughs. Its foundations can be traced back to the very beginnings of AI research in the 1950s, when pioneers like Alan Turing [18] raised the question of whether machines could ‘think’

in 1950, and when the Dartmouth Workshop [19] formally defined ‘Artificial Intelligence’ as a field in 1956. However, the notion of agents specifically began to crystallize in the 1980s, with work on symbolic planning, expert systems, and especially robotic architectures such as Ref. [20] that emphasized embodied, goal-driven behavior. By organizing the evolution of agentic AI into distinct phases (see Fig. 2), we provide a structured narrative of its development and identify trends that will shape its future.

2.1. The early foundations (1980s–1990s)

The concept of agentic AI emerged in the 1980s and 1990s, driven by foundational research in artificial intelligence, robotics, and cognitive science. The research published in Ref. [20] introduced behavior-based AI, emphasizing the importance of interaction with the environment over symbolic reasoning. This publication presents a layered control system that enables autonomous robots to operate with increasing competence. Its modular, hierarchical design allows for adaptive learning and goal-oriented decision-making. This work laid the groundwork for embodied and situated agents. In parallel, Aloimonos et al. [21] proposed active vision. The concept of Active vision reflects agentic AI principles, where intelligence emerges through interaction with the environment, and tasks such as navigation, manipulation, and learning are critical for the agent’s adaptability and goal-driven behavior. Shortly thereafter, Ref. [22] argued that intelligence could emerge from the interaction of simple behaviors rather than complex internal models, demonstrating a layered control system in which autonomous robots operate with increasing competence. It introduced a decentralized parallel architecture where independent activity producers interface with the real world, enabling autonomous decision-making and adaptability, key traits of agentic AI. Concurrently, researchers explored more formal characterizations of rational agents. The concept of rational agents was introduced by Ref. [23]. These rational agents act as entities that perceive and act to achieve goals. In the same year, Ref. [24] established the foundations of multi-agent systems (MAS), enabling collaborative and competitive agent interactions. It aligns with agentic AI by addressing the foundational principles for designing

Table 2
Agentic AI survey papers.

Title	Year	Area of focus	Specifications
Agentic AI in computer vision domain - Recent advances and prospects [9]	2023	Computer vision domain	This paper explores agentic AI in computer vision, covering its principles, technical innovations, applications, and evaluation methodologies.
The Landscape of emerging AI agent architectures for reasoning, planning, and tool calling: A survey [11]	2024	Single-agent and multi-agent architectures	This paper analyzes AI agent architectures, comparing single-agent and multi-agent systems in terms of design, communication strategies, and their impact on goal execution.
Agent AI: Surveying the horizons of multimodal interaction [4]	2024	Embodied multimodal agents	This survey is a comprehensive in nature defining agent AI for multimodal embodied systems. It covers agent design using foundation models for next-action prediction; discusses grounding to reduce hallucinations; applications in VR, robotics, and interactive environments. It provides a technical and implementation-focused perspective on embodied multimodal agents.
Large multimodal agents: A survey. [10]	2024	LLM-driven multimodal agents	This survey reviews large language multimodal agents (LMAs), covering their essential components, categorizing current research into four types, discussing collaborative frameworks for integrating multiple LMAs, compiling and standardizing evaluation methods, and highlighting their applications GUI automation, robotics, games, autonomous driving, video understanding and audio, visual generation & editing and future research directions.
AI agents under threat: A survey of key security challenges and future pathways [12]	2025	Security challenges	This survey reviews security challenges faced by AI agents, categorizing threats into four areas: unpredictability of multi-step user inputs, complexity in internal executions, variability of operational environments, and interactions with untrusted external entities. It highlights progress, limitations, and future research needed to develop more secure AI agent applications.
A survey of AI agent protocols [13]	2025	LLM-driven agents protocols	The primary focus of this paper is the analysis, classification, and future development of communication protocols for LLM agents.
The role of agentic AI in shaping a smart future: A systematic review [17]	2025	Role of agentic AI in organizational transformation	This review examines agentic AI's role in transforming organizations, discussing its attributes (autonomy, reactivity, proactivity, learning), applications in process automation and decision-making, the shift from Copilot to Autopilot models, strategic adoption frameworks, and challenges related to privacy, security, and ethics.
Agentic AI: Autonomous intelligence for complex goals—A comprehensive survey [6]	2025	Agentic-AI development methodologies and applications	This paper examines the theoretical foundations, key features, and technical approaches of Agentic AI, covering decision-making models, learning paradigms, and autonomy frameworks. It explores real-world applications across robotics, healthcare, business automation, and smart environments while addressing evaluation metrics and benchmarks. Additionally, it discusses ethical considerations and proposes frameworks for responsible integration and assessment of agentic AI systems.
Agentic AI for scientific discovery: A survey of progress, challenges, and future directions [14]	2025	Scientific discovery	This paper examines agentic AI in scientific research, highlighting its role in autonomous hypothesis generation, and data-driven discovery. It explores technical innovations, including machine learning frameworks and optimization methods, with applications in biological, physical, environmental, and social sciences.
Generative to agentic AI: Survey, conceptualization, and challenges [16]	2025	Transition from generative AI to agentic AI	This survey compares generative AI and agentic AI, discussing their definitions, capabilities, and differences. It analyzes key attributes, applications, and technical advancements enabling agentic AI and challenges toward AGI.
From large AI models to agentic AI: A tutorial on future intelligent communications [15]	2025	LAMs and agentic-AI technologies in intelligent 6G communication systems	This tutorial reviews the design, components, and applications of large artificial intelligence models (LAMs) and agentic AI in 6G communication systems. It offers a complete framework from foundational models to multi-agent coordination, along with future research challenges.
Agentic AI: The age of reasoning- A review (Ours)	2025	Structured review of agentic AI	This study explores the evolution of agentic AI, analyzes its core patterns, types, and operational environments while comparing several agentic AI systems. It highlights key advancements, challenges, and future research directions in developing autonomous, adaptable, and ethically aligned AI agents.

interactive and goal-driven systems that emphasize the importance of agent-environment interaction. Active vision systems moved beyond static image analysis, using camera movements and focus adjustments to gather relevant visual information.

In the domain of learning, Ref. [25] provided a rigorous mathematical framework for agentic behavior through reinforcement learning (RL), allowing agents to discover optimal policies through trial and error. Later in the same year, Ref. [26] introduced gradient-

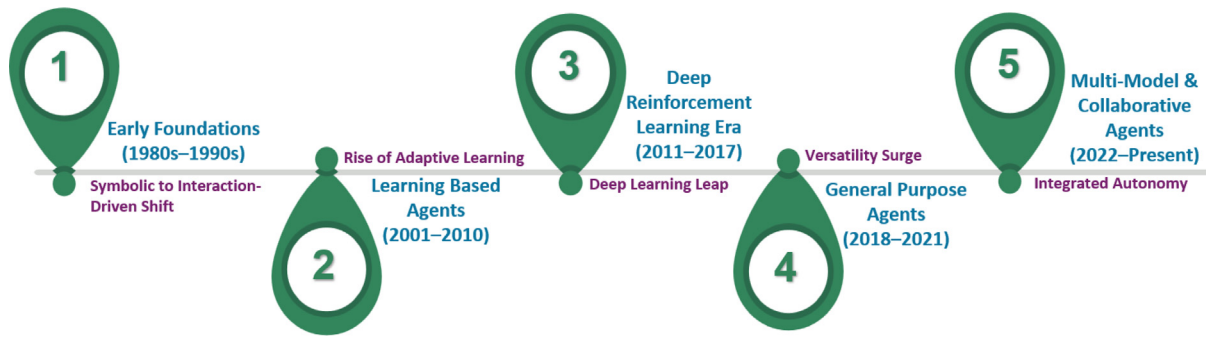


Fig. 2. Evolution of agentic AI: A timeline from symbolic foundations to multi-modal autonomy (1980s–present) – Tracing five phases with key milestones, the shift from symbolic to self-correcting intelligence.

based optimization for RL, enabling agents to learn in high-dimensional spaces. Ref. [27] demonstrated the power of inverse reinforcement learning (IRL), where agents could infer reward functions from observed behavior. This approach allowed agents to learn from human demonstrations, bridging the gap between human intent and machine decision-making. By the end of the 1990s, the field of agentic AI had established a robust theoretical and practical foundation, characterized by a shift from symbolic reasoning to interaction-driven intelligence. The emphasis on modularity, decentralization, and environmental interaction distinguished this era from earlier approaches to AI. However, while these advances enabled agents to operate autonomously in controlled environments, their scalability and adaptability to more complex real-world scenarios remained limited. Moreover, these early foundations established critical concepts that continue to influence contemporary agentic AI, including embodied cognition, rational agency, and learning through environmental interaction.

2.2. The rise of learning-based agents (2001–2010)

The 2000s saw machine learning techniques begin to reshape agentic AI, enabling more adaptive and scalable systems than ever before. Building on classic vision work, Ref. [28] introduced efficient algorithms for real-time object detection, allowing agents to perceive and track dynamic scenes. This cascade mechanism effectively acted as a focus-of-attention system, prioritizing computational resources based on environmental cues. Around the same time, Ref. [29] demonstrated the potential of RL in real-world applications, showcasing the ability of agentic AI to handle complex control tasks. Parallel progress in neural networks reignited by Ref. [30] set the stage for generative models to drive adaptive perception and decision making, where an agent refines its internal representations to classify and recognize environmental input more effectively. In the domain of multi-agent systems, Refs. [31,32] introduced novel algorithms for cooperative and competitive interactions among agents. Their work on multi-agent reinforcement learning (MARL) provided frameworks for agents to learn strategies in dynamic and uncertain environments, addressing challenges such as communication, coordination, and resource allocation.

Another critical contribution came from Ref. [33], who advanced the field of recurrent neural networks (RNNs) and long- and short-term memory (LSTM) networks. These architectures enabled agents to process sequential data and maintain memory over time, which was essential for tasks such as natural language processing, time-series prediction, and adaptive control. The rise of unsupervised and semi-supervised learning techniques further expanded the capabilities of agentic AI. For instance, Ref. [34] developed algorithms for learning representations from unlabeled data, allowing agents to generalize better between tasks and environments. This work laid the groundwork for transfer learning, where knowledge acquired in one domain could be applied to another, significantly improving the efficiency and scalability of agentic AI systems. Probabilistic reasoning also saw significant

advancements, with Ref. [35] proposing probabilistic graphical models for reasoning under uncertainty. These models allowed agents to make informed decisions in partially observable environments, enhancing their ability to handle real-world complexity. In this era, the advancement of learning-based agents, driven by reinforcement learning (RL), neural networks, and multiagent systems, enabled more adaptive and scalable AI decision making. However, challenges in interpretability, scalability, and efficiency persisted. This phase established learning as a central mechanism for agent adaptation and demonstrated the potential of data-driven approaches to overcome limitations of purely rule-based systems.

2.3. The deep reinforcement learning era (2011–2017)

The years 2011–2017 marked a turning point with the advent of deep reinforcement learning (DRL), which combined deep learning with RL to tackle past issues. Ref. [36] introduced the deep Q-network (DQN), which demonstrated that agents could learn to play video games directly from raw pixel data. This shift represented a leap toward more general agents. The same team [37] worked again and showed that DRL could achieve human-level performance in a wide range of Atari games, solidifying DRL as a cornerstone of agentic AI. The authors integrated deep learning with the reinforcement learning paradigm, enabling a single agent to achieve impressive performance on a variety of tasks purely from raw sensory inputs. This set the stage for the subsequent growth and sophistication of deep RL techniques, demonstrating how hierarchical perception plus reinforcement learning can yield powerful, highly adaptive agents. Ref. [38] showcased agentic AI's ability to master complex strategic games by defeating the world champion. Refs. [39,40] exemplified agentic AI by autonomously learning, adapting, and selecting actions through reinforcement learning and Tree Search in the complex Go environment without human knowledge and achieved superhuman performance through self-play, highlighting the potential of self-improving agents. Ref. [41] enabled agents to adapt quickly to new tasks with minimal data, a critical capability for real-world applications. This model-agnostic meta-learning algorithm endows an agent's parameters with rapid adaptability, requiring minimal data to achieve strong performance in classification, regression, and reinforcement learning tasks. Refs. [42,43] have gained attention as researchers prioritize the alignment of agentic AI with human values. By integrating human preference feedback as a reward specification [44], a key method to align agent objectives with human guidance, enabled advanced agentic RL behaviors in complex environments. This period marked the rise of deep reinforcement learning as a dominant strategy for equipping agentic systems with the ability to learn and evolve through interaction with their environment.

2.4. The emergence of general-purpose agents (2018–2021)

Between 2018 and 2021, the AI landscape underwent a remarkable shift toward agentic systems capable of increasingly autonomous reasoning, planning, and decision making. During this period, AI research converged on the idea of general-purpose agent systems designed to learn and adapt to a wide range of tasks, minimizing domain-specific engineering. A key precursor to this shift was the introduction of BERT [45], which popularized large-scale bidirectional pretraining for language understanding. Although BERT itself was not agentic, its success in fine-tuning across numerous NLP tasks paved the way for more goal-directed capabilities in subsequent language models. The introduction of GPT-2 [46] demonstrated large-scale generative language modeling with surprising zero-shot capabilities, highlighting a new level of autonomy for natural language processing tasks. Furthermore, GPT-2's ability to leverage large-scale unsupervised pretraining on the WebText corpus for emergent zero-shot task performance underscores how scaling model capacity can yield more autonomous, adaptable behavior across diverse NLP tasks without relying on explicit task-specific supervision. Ref. [47] demonstrated multi-agent teamwork and marked a leap toward collaborative agentic intelligence. Soon after, Ref. [48] introduced GPT-3, illustrating the next wave of general-purpose language models with drastically larger parameters and few-shot learning capabilities. This shift toward massive LLMs hinted at how agentic systems might integrate reasoning and action across numerous domains. Ref. [48] demonstrated that large language models (LLMs) could exhibit agentic behavior by generating goal-directed text and interacting with users. Ref. [49] experiments in Emergent Tool Use (OpenAI, 2020) within multiagent autocurricula showcased how agents might organically develop complex skills like tool usage when set loose in an evolving environment. By focusing on the single objective, the authors [50] suggested that reinforcement learning (RL) agents could develop the abilities necessary for artificial general intelligence, offering a unified and potentially more scalable approach to building intelligent systems. These influential works continued weaving the narrative of AI transitioning from task-specific solutions to agentic general-purpose systems, with deeper context awareness, emergent cooperation, and the ability to plan, act, and self-improve across diverse domains.

2.5. The current era: Multi-modal and collaborative agents (2022–present)

Multi-modal agents are deep learning systems that process text, images, audio, and more within a unified framework. They integrate techniques from NLP, computer vision, and speech to simulate human-like perception. These systems excel in tasks like image classification, visual QA, and text-to-image generation [51]. Most current systems focus on text-image integration, using various modality-conditioning approaches. Building on these capabilities, a growing body of research is now exploring how such systems can evolve into autonomous agents with reasoning and decision-making abilities. Recent advancements have focused more on multimodal agents, human-AI collaboration, and ethical considerations. Research by OpenAI, Google DeepMind, and others has explored the use of LLMs as autonomous agents capable of planning, reasoning, and executing tasks in dynamic environments. Ref. [52] showcases a single transformer-based policy model that can handle multiple domains (vision, language, robotic control), hinting at general 'agentic' behavior. The study [53] highlighted the impact of scale in agentic AI, demonstrating that larger models can autonomously generalize tasks with minimal supervision, achieving breakthrough performance in reasoning, multilingual processing, and code generation.

With the advent of ChatGPT 4 [54] and its variant, the study illustrated how massive model scaling, alignment strategies, and user-oriented interfaces can yield further emergent abilities and elevated performance, ultimately accelerating the path toward more agentic

AI systems. This trend has accelerated with projects like Hugging-GPT [55], where a single coordinating language model orchestrates a wide range of specialized models to tackle complex, multi-step tasks, effectively acting as a meta-agent. BLIP-2 [56] efficiently aligns vision and language modalities by combining frozen image encoders with large language models. Meanwhile, Kosmos-1 and 2 [57,58] integrate grounded text generation and visual understanding in a single framework. Kosmos-2 [58] extends Kosmos-1 [57] by incorporating grounded object detection and multimodal understanding, expanding the scope of multimodal reasoning. Similarly, Flamingo [59] and SPORT [60] extend few-shot visual QA capabilities and tool usage integration, respectively, showing how multi-modal agents can perform perception-grounded decision-making with minimal task-specific tuning. RoboTwin 2.0 [61] further exemplifies these capabilities in multi-robot systems using vision-language foundation models for coordinated manipulation and navigation.

More recently, studies like "Generative agents: Interactive Simulacra of Human Behavior" [62] have showcased how language-based models can evolve into 'generative agents', displaying emergent social and cognitive patterns. By incorporating persistent memory, planning, and reflection, these agents simulate human-like behaviors in shared virtual environments, highlighting a new level of autonomy and adaptability. Meanwhile, PaLM-E [63] extends these agentic capabilities to the physical domain by uniting vision, language, and action in robotics tasks. This embodied approach exemplifies how LLMs can be integrated with real-world sensors and actuators to navigate and manipulate their surroundings, a crucial step toward robust, embodied agents. Another line of research tackles the challenge of making LLM-driven agents more reliable and self-corrective. RoboMind [64] further demonstrates these capabilities in multi-robot systems by leveraging vision-language foundation models for coordinated manipulation and navigation.

Collaborative agents represent an equally transformative direction in agentic AI by enabling multiple specialized agents to coordinate, communicate, and solve complex tasks collectively. Frameworks like AutoGen [65], LangGraph [66], and CrewAI [67] provide structured approaches to define agent roles, communication protocols, and workflows, allowing agent teams to decompose problems, allocate subtasks efficiently, and integrate output seamlessly. Collaborative LLM agents for web navigation [68] demonstrate how such systems can dynamically navigate, extract, and integrate web-based information by dividing tasks among agents. The ReAct framework [69] merges reasoning through chain-of-thought with the usage of the acting tool and environmental steps, enabling language models to parse a scenario, plan actions, and adapt as they receive feedback. Similarly, reflexion [70] introduces an iterative self-reflection mechanism, allowing agents to identify errors or dead ends in their outputs and revise their strategies.

Beyond academic papers, open-source efforts such as AutoGPT and BabyAGI (2023–present) have rapidly popularized the notion of chaining LLMs together with planning loops and external APIs to perform tasks with minimal human supervision. These community-driven projects highlight the democratization of agentic AI, showing how relatively straightforward pipelines can endow an LLM with memory, a sense of goals, and the ability to decompose complex tasks into actionable steps. Into this tapestry come DeepSeek models, such as the frameworks proposed by Refs. [71–73] and expanded by Ref. [74]. Their approach emphasizes retrieval-augmented reasoning: the AI 'seeks' relevant context from massive knowledge sources, then funnels that data into adaptive planning loops. This capability not only frees AI from static knowledge constraints, but also pushes it closer to genuine self-directed problem solving, paving the way for longer-term, flexible, and goal-oriented behavior in dynamic environments. Moreover, federated learning (FL) also enables collaboration among multiple autonomous agents without exposing their raw data. For example, FedMarL [75] integrates FL with multi-agent reinforcement learning (MARL), where agents work together to optimize client selection, achieving higher model accuracy, reduced latency, and lower communication costs. In

another study [76], FL allows robots to learn socially appropriate behaviors within their local environments while sharing insights to strengthen the group's overall social intelligence. MAF-DRL [77] empowers autonomous agents in wireless sensor networks (WSNs) to collaboratively detect attacks using Federated Learning and Multi-Agent Reinforcement Learning. A vertical federated multi-agent reinforcement learning framework [78] enables dynamic interaction between attacker and defender agents. By focusing on DQN in stationary and A2C in nonstationary environments, the system provides adaptive, privacy-preserving policy learning in federated settings. While Federated Learning combined with Multi-Agent Systems or Reinforcement Learning promises adaptive, privacy preserving collaboration among distributed agents, current applications remain domain specific and often focus on narrow optimizations. For instance, stability in spiking neural networks [79], wearable health data management [80], autonomous driving with preference-based multi-agent RL [81], and satellite edge computing resource allocation [82] highlight the need for generalizable frameworks that unify these approaches under broader agentic AI objectives and alignment.

Taken together, these developments indicate that AI is steadily moving from passive NLP models toward fully agentic systems capable of reasoning, planning, adapting, and even self-correcting in response to novel challenges. As research continues to refine memory mechanisms, alignment strategies, and multimodal integration, we can expect agentic AI to push further into domains demanding human-like creativity, judgment, and autonomy, ultimately charting the path to more general, self-improving, and context-aware intelligence.

3. Key aspects of agentic AI

3.1. Patterns of agentic AI

Agentic AI patterns are design patterns or frameworks used to structure and implement agentic behavior in AI systems. These patterns explain how agentic AI systems interact, process tasks, and respond to user input. It focuses on structural methodologies for building systems (agents). The researchers [3,83] explored four basic patterns of agentic AI frameworks: tool use, allowing external resource integration; reflection, fostering iterative self-correction; planning, supporting structured task decomposition; and multi-agent collaboration (MAC), coordinating multiple LMs for specialized sub-tasks. Table 3 comprehensively describes the different patterns of agentic AI.

3.2. A taxonomic framework to classify agentic AI

Based on our systematic review, this study proposes a progressive taxonomic framework to classify agentic AI systems. Fig. 3 provides a hierarchical taxonomy of Agentic AI types, underscoring the gradual advancement in autonomy, adaptability, and cognitive complexity from reactive agents to self-aware systems. The following are agentic AI categorized into several distinct types based on capabilities, decision-making autonomy, and learning mechanisms.

3.2.1. Reactive agents

Reactive agents are based on current input without memory or previous experiences [84]. These agents follow predefined rules or heuristics and cannot adapt their behavior dynamically. They are efficient for well-defined tasks, but lack learning capabilities. Brooks [20] and Mataric [85] emphasized simplicity through modular designs, where complex behaviors are decomposed into simpler, independent modules. This modularity not only facilitates rapid response to changes in the environment but also simplifies the implementation in both single and multi-robot scenarios.

Conversely, while Arkin [86] acknowledged the benefits of such simplicity, his work delved into more intricate aspects of reactive

control, including conflict resolution between behaviors and the integration of reactive layers with higher-level planning when necessary. Ref. [87] examined that reactive agents respond instantly to stimuli based on predefined rules. Later, Ref. [88] extended the discussion to distributed systems, highlighting the nuances that arise when reactive agents are deployed in a decentralized, networked environment. Simple Reflex Agents [89], Behavior-Based Agents [90], Subsumption Architecture Agents [91], Finite-State Machine (FSM) Reactive Agents [92], Hybrid Reactive Agents [93] are some examples of reactive agents.

3.2.2. Proactive agents

Proactive agents anticipate future events and take the initiative to act before explicit instructions are given. These agents analyze patterns, predict needs, and make autonomous decisions to optimize outcomes [94]. By integrating predictive models, machine learning techniques, and advanced decision-making frameworks, proactive agents are poised to address challenges in fields ranging from autonomous vehicles and smart grids to healthcare and industrial automation. Proactive agents can dynamically adjust their behavior based on feedback from the environment, leading to improved anticipatory performance over time [95–97]. Types of proactive AI agents, such as personal assistants [98] and predictive maintenance [99,100] tools, predict and address issues autonomously, optimize operations, and reduce costs in sectors like technology, healthcare, and manufacturing. Table 4 provides a comparison of reactive and proactive agent approaches, which outline their behaviors, advantages, and challenges.

3.2.3. Limited memory agents

These agents retain historical data for a short duration, allowing them to improve decision making by analyzing past inputs [101]. Limited memory agents analyze past interactions within a defined time frame to improve their adaptability and predictive capabilities [102]. This ability is vital in domains such as autonomous driving, healthcare diagnostics, and predictive maintenance. One of the core components of limited-memory agents is the memory module, which allows AI models to retain and process previous input to generate more informed decisions. Recent advancements in AI, particularly in large language models (LLMs), have significantly improved the efficiency of memory mechanisms in AI agents. A study by Ref. [103] highlights that memory sharing frameworks in LLM-based agents enhance their ability to recall past interactions, thus improving their contextual understanding and response accuracy. The ability to store and retrieve episodic data has made these agents more robust in handling dynamic tasks, such as dialogue-based AI systems and personalized recommendations.

Historically, AI systems lacked the ability to retain memory beyond the immediate computation cycle. However, research in memory augmentation has introduced methodologies where AI agents can selectively store and retrieve relevant information, enabling more sophisticated learning strategies. For instance, a proposal for intelligent agents with episodic memory by Ref. [104] discusses how structured memory storage mechanisms can significantly enhance decision-making efficiency. Such memory-augmented AI systems can effectively learn from past interactions without requiring extensive retraining, making them more scalable and efficient in real-world applications. Future limited memory AI agents will integrate hybrid memory models with transformer-based architectures to enhance reasoning and expand applications in finance, education, and robotics [105]. Bayesian Agents [106], Recurrent Neural Network (RNN)-Based Agents [107], Long Short-Term Memory (LSTM) Agents [108], Transformer-Based Agents [109] and Chatbots and Conversational AI Agents [110] are some types of LMA (Limited Memory Agents). Moreover, limited-memory AI agents struggle to balance memory efficiency with decision-making while scaling to high-dimensional environments and retaining context in real-time. Fig. 4 shows the significance of the memory module in agents.

Table 3
Comprehensive overview of patterns of agentic AI.

Name	Description	Key features	Technical implementation	Key models used	Performance metrics	Real-world applications	Scalability	Ethical considerations	Cost implications
Reflection Pattern	Evaluates initial responses and refines them iteratively for better outcomes.	Self-evaluation, iterative refinement of outputs.	Feedback loop with self-critique mechanisms, error analysis.	Transformers, self-supervised learning.	Accuracy of iterations, convergence speed.	Self-correcting chatbots, creative AI, and debugging systems.	Scales well with additional computational resources but requires optimization for large tasks.	May introduce bias if feedback mechanisms are flawed.	Medium computational cost due to repeated iterations.
Tool Use Pattern	Leverages external tools or APIs to fetch or compute results.	Interaction with tools, integration with APIs.	API calls, vector search engines, and integration layers.	GraphQL APIs, vector search models.	Latency, retrieval accuracy.	Knowledge retrieval, advanced computations, search engines.	Highly scalable depending on the tool infrastructure.	Dependency on external tools may raise transparency concerns.	Low operational cost; tool dependency reduces overhead.
ReAct Pattern	Combines reasoning with action to address queries dynamically.	Real-time reasoning, action execution, environment interaction.	Logic-action alternation mechanisms, action validation.	Reinforcement learning, decision trees.	Task completion time, reasoning accuracy.	Robotics, autonomous driving, and interactive problem-solving systems.	Scales depending on the complexity of reasoning and task.	Risk of error propagation in high-stakes tasks.	High cost for real-time systems; significant computation required.
Planning Pattern	Solves complex problems through multi-step strategies.	Goal-oriented planning, iterative execution, refinement.	Sequence modeling, task division into subtasks.	Markov decision processes, hierarchical models.	Efficiency of task breakdown, task success rate.	AI-driven project planning, strategic simulations, and workflow automation.	Scales well but may become slower for deeply nested tasks.	Misaligned goals may introduce ethical concerns in critical applications.	Moderate computational cost due to step-by-step refinement.
Multi-Agent Pattern	Employs collaboration among multiple AI agents for problem-solving.	Distributed processing, modularity, specialized agents.	Communication protocols between agents, task assignment.	Multi-agent reinforcement learning.	Coordination success rate, execution time.	Swarm robotics, simulations, modular AI systems.	Highly scalable for parallelizable tasks but complex coordination.	Risk of unintended behavior from agent conflicts.	High operational cost due to communication overhead.

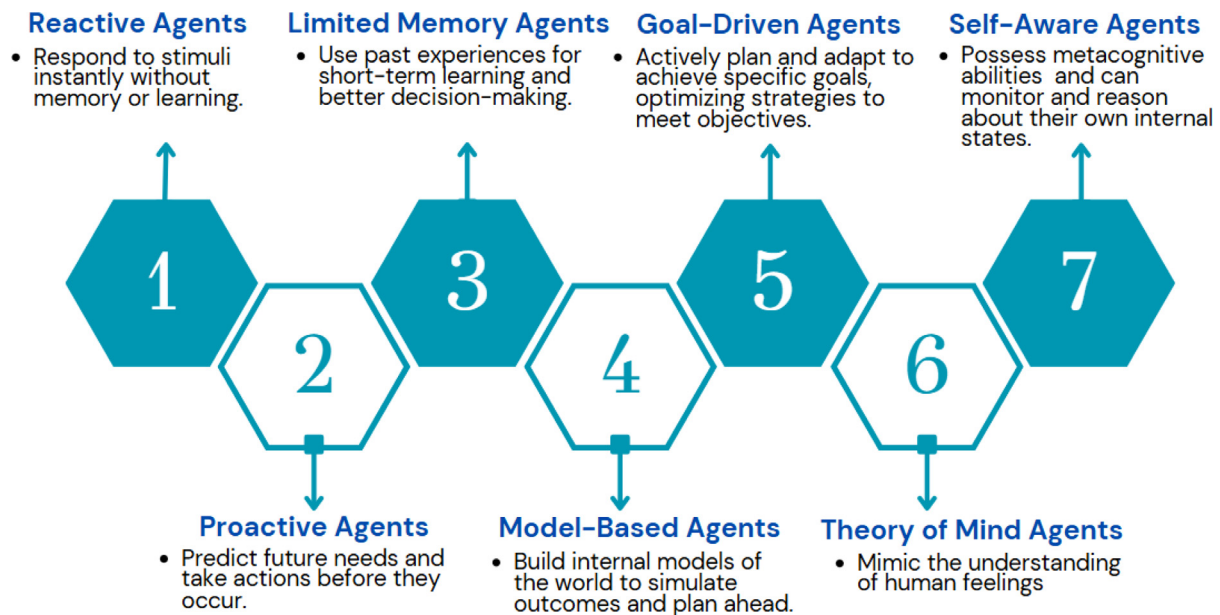


Fig. 3. Progressive taxonomy of agentic AI types from reactive to self-aware systems.

3.2.4. Model-based agents

Model-based agents maintain structured representations of their environments, allowing them to predict and plan future actions [38]. They rely on environmental modeling to construct and maintain an internal representation of their surroundings, enabling accurate predictions and informed planning [38]. Through predictive planning, these agents

simulate potential outcomes before executing actions, ensuring that decisions are based on anticipated rather than immediate responses [112]. Their adaptability allows them to continuously refine their internal models based on new information, improving long-term performance in dynamic environments [113]. However, these advantages come at the cost of being computationally intensive, as the need for simulations and

Table 4
Comparison of proactive, reactive, and hybrid AI agents.

Aspect	Proactive AI agents	Reactive AI agents	Hybrid approach
Definition	Anticipate needs and act autonomously based on predictions and strategic data analysis, addressing issues before they occur.	Respond to events as they occur, making decisions in real-time based solely on current environmental inputs.	Combines proactive forecasting with reactive responsiveness to handle immediate issues while planning for future challenges.
Advantages	Enhance efficiency and foresight, prevent problems, reduce downtime, and optimize processes by anticipating changes.	Provide immediate responses in dynamic situations, ensuring timely handling of issues with rapid, real-time reactions.	Balances short-term responsiveness with long-term planning, optimizing both immediate outcomes and future performance.
Use cases	IT management (predicting system failures and autonomously initiating maintenance), predictive maintenance in manufacturing.	Customer service chatbots that handle inquiries in real-time; safety systems in autonomous vehicles reacting instantly to road conditions and obstacles.	Smart home systems that react to user commands while learning habits to proactively adjust settings for comfort and energy efficiency.
Challenges/Considerations	Requires sophisticated predictive models, comprehensive high-quality data, and advanced machine learning techniques, leading to greater technical complexity.	Must be robust and fast to process real-time data effectively, even though they are generally simpler than proactive systems.	Involves complex integration, balancing technical, practical, and ethical considerations to merge immediate reactivity with long-term proactive planning efficiently.

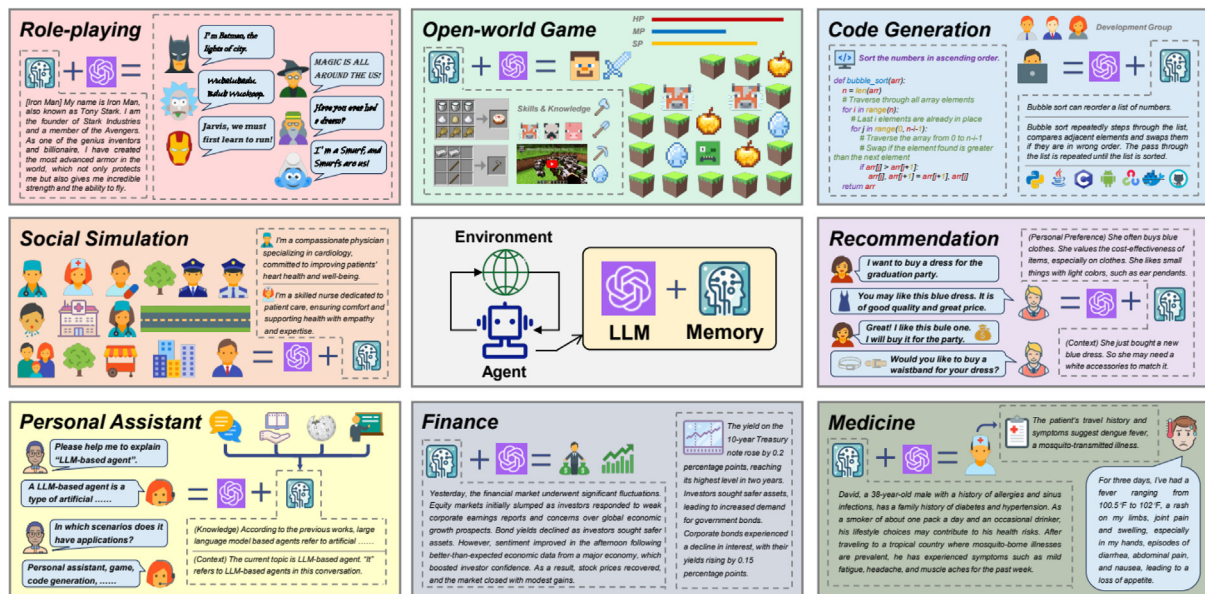


Fig. 4. The significance of the memory module in agents powered by large language models (LLMs) [111].

planning increases resource demands [114]. Despite this, model-based agents demonstrate improved sample efficiency, requiring fewer interactions with their environment compared to model-free approaches, making them particularly valuable in domains where data collection is costly and limited [115].

Early work in model-based AI focused on symbolic reasoning, exemplified by the Dyna architecture [116], which integrated planning with reinforcement learning (RL). Modern advances have shifted toward data-driven approaches, such as PILCO [117], which combines probabilistic models with policy search, and world models [118], which use deep learning to simulate environment dynamics. Recent trends emphasize hybrid models that merge learned representations with symbolic reasoning for better generalization [119]. Model-based agents face challenges such as error propagation in environment modeling, high computational demands for real-time planning in complex state spaces, and the need for extensive interactions to learn accurate models, limiting scalability. Model-based agents can be categorized into three primary types: Symbolic Model-Based Agents [120,121], Probabilistic Model-Based Agents [122] and Learned Model-Based Agents [123,124]. Table 5 illustrates different learning paradigms for model-based agentic AI across various components.

3.2.5. Goal-driven agents

Goal-driven agents are based on predefined goals, optimizing their strategies to achieve desired results [125]. These agents operate within a framework where their behavior is directed toward satisfying specific goals, often through planning, reasoning, and adaptive decision making. Key characteristics include goal-oriented autonomy, strategic planning, and utility maximization, enabling them to prioritize tasks, allocate resources, and adjust actions in response to environmental feedback. Early research in goal-driven AI emerged from classical planning systems, such as STRIPS [126] and SOAR [127], which focused on symbolic goal decomposition. Goal-driven agents can be categorized based on their operational frameworks: Simple Goal-Based Agents [128], Utility-Based Agents [129], Learning Agents [130], and Multi-Agent Systems [131]. Key challenges for goal-driven agents include dynamic goal adaptation in changing environments, scalability in high-dimensional state spaces, ensuring goal alignment with human values in safety-critical domains, and achieving objectives despite partial observability and noisy data.

3.2.6. Theory of mind (ToM) agents

Theory of mind (ToM) agents can infer human emotions, beliefs, and intentions, making them more effective in human-centered environments [132]. These agents leverage cognitive and social reasoning to

Table 5
Learning paradigm of model-based agentic AI.

Component	Function	Learning Approach	Purpose
Model learning	Learns environment dynamics.	– Supervised – Unsupervised – Reinforcement Learning.	Simulate outcomes for planning.
Policy learning	Determines the best action to take.	Reinforcement Learning (with or without planning).	Goal achievement through optimal behavior.
Value function learning	Estimates long-term reward of actions.	Temporal-Difference, Monte Carlo methods.	Guides policy and planning decisions.
Self-supervised adaptation	Continuously refines internal model using prediction errors.	Self-supervised learning.	Enhances model accuracy and adaptability over time.

predict human behavior, enabling more natural collaboration, personalized assistance, and empathetic responses. The main characteristics include mental state attribution, context-aware adaptation, and social cognition, allowing them to bridge the gap between mechanistic AI operations and human-like social understanding [133–136]. The concept of ToM originated in developmental psychology [137], later adopted in AI for social robotics and human–computer interaction. Early work by Ref. [138] demonstrated ToM in robots for gaze-following and intention recognition. Modern advancements include Machine Theory of Mind [133], which uses deep learning to infer mental states from observational data, and social NLP models like GPT-4 [54], which simulate empathy in dialogue. ToM agents are implemented through cognitive architectures [139,140] (symbolic reasoning like BDI), machine learning [141], and hybrid models [142].

The main challenges in developing agentic AI for theory of mind (ToM) include handling the dynamism and unpredictability of real-world environments, which affect agents' beliefs and decision-making processes [143–146]. Cognitive modeling difficulties arise in balancing explicit and implicit belief representation, integrating emotional and cognitive theories, and addressing the inadequacy of existing mathematical frameworks [147–149]. Scalability remains a key issue, as considering multiple factors for ToM inference is computationally challenging [150,151]. Social interaction complexities include determining initial interactions, crowdsourced knowledge evaluation, real-time human gesture recognition, and implementing social capabilities in humanoid robots [152,153].

3.2.7. Self-aware agents

Self-aware agents possess metacognitive abilities, which allow them to reason about their internal states and adapt autonomously [154]. These agents transcend task-specific functionality by maintaining awareness of their knowledge, limitations, and decision-making processes, allowing them to self-optimize, recover from failures, and explain their actions [155]. The main characteristics are introspection, self-monitoring, adaptive learning, and autonomous goal refinement, distinguishing them from traditional AI systems that operate without self-referential awareness. Self-aware AI emerged from cognitive architectures like SOAR [127] and ACT-R [156], which incorporated limited self-monitoring for problem-solving. Modern advancements include conscious AI frameworks [157], which model self-awareness through global workspace theory, and self-reflective neural networks [158], which use meta-learning to adapt their parameters dynamically.

Contemporary applications span autonomous robotics (self-diagnosing drones), AI safety (systems that detect algorithmic bias), and artificial general intelligence (AGI) [154] research. Recent breakthroughs, such as self-aware reinforcement learning [159], demonstrate agents capable of reconfiguring their policies in response to performance feedback. Self-aware agents are typically categorized as symbolic (employing formal logic and rule-based systems), neural (utilizing deep learning architectures, such as RNNs and transformers for emergent metacognition), or hybrid (integrating symbolic reasoning with neural components for robust self-awareness). These agents encounter significant scalability challenges where complex environments demand computationally expensive self modeling along

with explainability hurdles stemming from black-box architectures that limit interpretability and robustness concerns that necessitate rigorous validation and ethical considerations for reliability in safety-critical applications [159].

Table 6 compares different types of agentic AI, highlighting their core aspects. It also contrasts these types to emphasize their distinctions.

3.3. Capabilities and limitations of agentic systems

This systematic review identified key capabilities and limitations across the agent types in our taxonomic framework. This section analyzes these dimensions, highlighting both the progressive enhancement of capabilities and the persistent challenges that remain.

3.3.1. Perception capabilities

Perception capabilities have evolved significantly across agent generations, from simple sensor processing in reactive agents to sophisticated multi-modal understanding in contemporary systems.

- **Reactive and Proactive Agents:** Limited to predefined feature extraction from structured inputs, typically operating on low-dimensional sensor data or symbolic representations. These agents struggle with ambiguity, noise, and novel perceptual patterns.
- **Limited Memory and Model-Based Agents** incorporate temporal context and learned feature representations, enabling more robust perception in noisy environments. These agents can recognize patterns across time and adapt perceptual processing based on experience.
- **Goal-Driven and Theory of Mind Agents** demonstrate context-aware perception, selectively attending to task-relevant features and integrating multi-modal inputs. These agents can interpret perceptual information in relation to goals and social context.
- **Self-Aware Agents** exhibit metacognitive awareness of perceptual limitations, enabling uncertainty estimation and active information seeking to resolve ambiguities. These agents can reason about what they do not perceive and adjust accordingly.

Despite these advances, significant challenges remain in developing perceptual systems that match human capabilities in understanding novel environments, recognizing unusual situations, and integrating information across modalities under resource constraints. These limitations are caused by three factors: first, perception models are trained on static datasets that fail to generalize to novel situations and environments due to distribution mismatch; second, lack of representations for rare or unusual situations like safety-critical scenarios; third, agents perceive data from multiple sensory sources and surroundings that process text, images, audio, etc. However, LLMs often misinterpret sensory data due to lack of grounding, leading to hallucinations or incorrect assumptions. They also struggle with unfamiliar inputs that were not seen during training.

Perception is the first stage of agentic systems. If an error occurs at this stage, it will propagate downstream. For instance, misperception of a situation can lead to erroneous planning and conclusions.

Table 6

A comparative analysis of goal-driven agents, theory of mind (ToM) agents, and self-aware agents, highlighting their differences across core aspects.

Aspect	Goal-driven agents	Theory of mind (ToM) agents	Self-aware agents
Primary objective	Achieve predefined goals through optimization and strategic planning.	Infer human mental states (emotions, beliefs, intentions) for collaboration.	Introspect, adapt, and self-optimize by reasoning about internal states.
Core features	– Goal-oriented autonomy – Utility maximization – Strategic planning.	– Mental state attribution – Social cognition – Context-aware adaptation.	– Metacognition – Self-monitoring – Autonomous goal refinement.
Key technologies	Classical planning (STRIPS, SOAR), reinforcement learning, multi-agent systems.	Cognitive architectures (BDI), deep learning (Machine ToM), NLP (GPT-4).	Meta-learning, self-reflective neural networks, hybrid symbolic-neural frameworks.
Applications	Autonomous logistics, healthcare optimization, enterprise automation.	Social robotics, personalized education, empathetic customer service.	AGI research, self-diagnosing drones, AI safety (bias detection).
Challenges	– Dynamic goal adaptation – Scalability in state spaces – Goal alignment.	– Unpredictable human behavior – Social interaction complexity – Scalability.	– Scalable self-modeling – Black-box explainability – Robustness validation.
Ethical considerations	Aligning AI goals with human values (like safety-critical domains).	Privacy in mental state inference, ethical use of empathy simulation.	Accountability for autonomous decisions, transparency in self-aware reasoning.
Future directions	Hybrid human-AI goal-setting frameworks, scalable planning in dynamic environments.	Hybrid symbolic-ML models for social reasoning, real-time gesture recognition.	Explainable self-aware architectures, ethical frameworks for AGI.

This puts the agentic system's reliability into question and at stake. Some research studies have previously recommended domain randomization [160] and sim-to-real transfer [161] approaches that inject variability during training. However, these approaches cause the model to learn overly generic behaviors that underperform in real-world contexts and have limited scope, meaning that they cannot cover the wide distribution of situations encountered in agentic AI systems. In recent approaches, the active perception [162,163] mechanism has shown effectiveness in reducing reliance on brittle passive sensing by allowing the system to dynamically reconfigure its sensory inputs based on task context. Recent advances in multimodal pretraining, such as BEiT-3 [164] and Flamingo [59], have improved alignment between language and perception. However, current models still struggle in open-set environments, and their high resource demands limit deployment on low-power edge devices.

To address these challenges, one approach is to integrate agentic systems with grounded perception systems (computer vision and sensor fusion models) trained on real-world sensor data, resulting in more reliable and factually accurate understanding. Another strategy involves developing multi-modal foundation models for deeper contextual awareness, along with implementing anomaly detection mechanisms to manage unexpected inputs through clarification or human assistance.

3.3.2. Memory and knowledge representation

Memory architectures have evolved from the absence of state in reactive agents to sophisticated multi-component systems in advanced agent architectures.

- Reactive Agents operate without persistent memory, processing only current inputs without historical context.
- Proactive and Limited Memory Agents maintain simple historical records and statistical summaries, enabling pattern recognition and basic adaptation.
- Model-Based and Goal-Driven Agents implement structured knowledge representations and episodic memory systems, supporting simulation, planning, and contextual retrieval.
- Theory of Mind and Self-Aware Agents incorporate declarative, procedural, episodic, and semantic memory components with sophisticated retrieval mechanisms, enabling complex reasoning and knowledge transfer.

Contemporary LLM-based agents have demonstrated particularly interesting memory architectures, combining:

1. **Parametric memory:** Knowledge encoded in model weights during pretraining.
2. **Working memory:** Information maintained in context window during operation.
3. **External memory:** Retrieval-augmented approaches accessing document stores, vector databases, or tool outputs.

These hybrid approaches address limitations of pure parametric knowledge, including hallucination tendencies, knowledge cut-offs, and context window constraints. However, challenges remain in developing memory systems that efficiently balance comprehensiveness with retrieval precision and integrate seamlessly with reasoning processes. Agentic AI systems rely heavily on high-fidelity memory to support tracking goals, plan over time, and engage in multistep conversations. However, current memory types have key flaws. Parametric memory (inside the model) cannot be updated, leading to outdated knowledge. Working memory is limited in size, making it hard to handle long tasks. External memory systems, like databases, often return irrelevant or poorly ranked information. These gaps cause AI to forget, lose task focus, or make incorrect decisions. A better dynamic memory system is needed: one that can intelligently store, forget, and retrieve context based on the task at hand. Recent solutions aim to fix memory issues such as MemGPT [165] introduces a modular system that separates short-term, long-term, and episodic memory, helping agents recall key experiences across tasks. ReLoRA [166] and Streaming Transformers [167] enhance working memory by enabling longer context handling and reducing processing limits. Models like Retro [168], REALM [169], and KNN-LM [170] improve retrieval accuracy by using training signals to reduce irrelevant results. Latent cache distillation [171] integrates frequently retrieved knowledge back into the model's internal memory for better grounding. A promising direction is agent-level memory policies [172], where AI learns what to store, when to retrieve it, and how to apply it in reasoning. These advances move toward smarter, more context-aware systems. However, it remains a challenge for researchers.

Future agentic AI systems may incorporate centralized memory controllers that dynamically coordinate between parametric, working, and external memory, optimizing retrieval and storage based on task relevance and salience. Memory modules could become self-adaptive, learning to manage retention and forgetting through reinforcement learning, thereby enabling efficient memory use across diverse tasks.

3.3.3. Reasoning and decision-making

Reasoning capabilities show perhaps the most dramatic progression across our taxonomic framework, from simple condition-action mappings to sophisticated multistep reasoning processes.

- **Reactive and Proactive Agents:** Implement direct mappings from situations to actions, with limited or no explicit reasoning processes. Decision making is typically based on predefined rules or simple pattern matching.
- **Limited Memory and Model-Based Agents:** Demonstrate basic inferential capabilities, including statistical reasoning and simple causal modeling. These agents can evaluate alternative actions through simulation, but typically rely on domain-specific heuristics.
- **Goal-Driven and Theory of Mind Agents:** Exhibit more sophisticated planning and counterfactual reasoning, including multistep action sequences and consideration of others' perspectives. These agents can decompose complex problems and adapt reasoning strategies to different domains.
- **Self-Aware Agents:** Demonstrate metacognitive reasoning, including awareness of reasoning limitations, strategy selection, and self-correction. These agents can potentially engage in abstract reasoning across domains and generate novel problem solving approaches.

Recent advances in LLM-based agents have demonstrated particularly promising reasoning capabilities through techniques such as:

1. **Chain-of-thought prompting:** Eliciting step-by-step reasoning processes.
2. **Self-reflection:** Evaluating and refining reasoning attempts.
3. **Tool use:** Augmenting reasoning with external computational resources.
4. **Ensemble methods:** Combining multiple reasoning approaches.

Despite these advances, significant limitations remain in logical consistency, causal reasoning, and handling of complex, novel problem structures. Current systems still struggle with compositional generalization and often exhibit reasoning failures when faced with problems requiring multiple interdependent inferences. These limitations arise when agentic systems focus on predicting the next word without ensuring logical consistency or understanding cause-and-effect. As a result, they struggle with complex tasks that require explicit symbolic reasoning and causal structures. This leads to contradictory answers and missing dependencies, limiting their reliability in real-world applications where long-term planning and consistent reasoning are essential. To address these challenges, researchers are developing methods like self-consistency decoding [173] to improve logical answers and neuro-symbolic integration to reduce contradictions using rule-based checks. Causal reasoning models such as CausalQA [174,175] and causal inference [176] help to distinguish true cause–effect relationships, while structured reasoning frameworks like Chain-of-Thought (CoT) [177] Tree-of-Thought [178], ReAct [69], and AutoGPT [179] enable better reflection and decision making in complex tasks. Researchers also use Neuro-Symbolic AI by combining large language models (LLMs) with symbolic reasoners such as theorem provers [180] and constraint solvers [181]. Moreover, giving agents access to external tools such as code interpreters and calculators enables them to handle complex calculations and verify their own reasoning steps.

This shows that agentic AI systems are gradually moving toward building architectures that are structured for logical reasoning and decision-making. In the future, one possible solution can be to incorporate training objectives into the optimization process that integrate logical consistency and formal verification. This would help agentic systems internally validate their reasoning rather than rely on sampling-based corrections. Agentic AI systems may use a combination of structured reasoning techniques, such as chain-of-thought prompting and self-reflection, to cross-verify their outputs, reduce logical errors, and enhance decision-making reliability. These approaches are key to building more reliable and adaptable agentic AI systems.

3.3.4. Learning and adaptation

Learning mechanisms have evolved from the absence of adaptation in reactive agents to sophisticated multi-strategy approaches in advanced systems.

- **Reactive Agents** lack learning capabilities, with behavior fixed at design time.
- **Proactive and Limited Memory Agents** implement basic statistical learning and pattern recognition, enabling gradual adaptation to environmental regularities.
- **Model-Based and Goal-Driven Agents** demonstrate more sophisticated learning through model updating, reinforcement learning, and transfer across related tasks.
- **Theory of Mind and Self-Aware Agents** exhibit meta-learning capabilities, including learning how to learn, strategy adaptation, and curriculum self-generation.

Contemporary agent architectures often combine multiple learning mechanisms:

1. **Supervised learning:** For acquiring skills from demonstrations.
2. **Reinforcement learning:** For optimizing behavior through environmental feedback.
3. **Self-supervised learning:** For representation learning from unlabeled data.
4. **Few-shot learning:** For rapid adaptation to new tasks with minimal examples.
5. **Continual learning:** For accumulating capabilities without catastrophic forgetting.

However, significant challenges remain in developing systems that can efficiently learn from limited data, transfer knowledge across dissimilar domains, and maintain performance across distribution shifts without extensive retraining. This is because the lack of lifelong learning and meta-learning capabilities makes agentic AI inefficient in dynamic and data-sparse conditions. This causes systems to struggle with limited supervision, unseen domains, or evolving tasks without retraining from scratch. To mitigate these problems, continual learning [182] enabled agent architectures update memory modules, skill libraries, and planners incrementally using techniques such as experience replay, regularization, or generative pseudo-memory. Additionally, Meta-agent frameworks supporting fast adaptation, such as Meta-World [183,184] and LEAP [185]; domain-aware perception and planning modules that detect shifts and adjust internal behavior, like AgentAda [186]; and agent ecosystems that share knowledge between tasks or deployments, such as federated agent learning [187] and skill sharing protocols [188], are extensively explored. But meta-agent systems often fail to generalize to structurally different tasks and lack effective long-term memory consolidation for skill transfer. Domain-aware perception and planning are based on fixed adaptation rules and struggle in non-stationary or partially observable settings. Agent ecosystems face difficulties in synchronizing knowledge, managing diverse architectures, and ensuring privacy and consistency during communication, underscoring the need for more autonomous and adaptive agentic systems.

In the future, exploration can focus on federated multi-agent reinforcement learning and swarm learning, where agents learn from their local environments, share distilled skills periodically, and independently validate incoming knowledge based on their own goals. This decentralized approach enables heterogeneous agents with varying sensors and roles to collaboratively update shared knowledge while maintaining autonomy and privacy, supporting scalable and adaptive agentic systems in dynamic, non-stationary environments.

3.3.5. Autonomy and initiative

Autonomy levels vary substantially across agent types, from the rigid behavior of reactive systems to the self-directed operation of advanced agents.

- Reactive and Proactive Agents demonstrate limited autonomy, operating within narrow predefined parameters with minimal initiative.
- Limited Memory and Model-Based Agents exhibit increased operational autonomy, adapting to environmental variations and selecting from alternative approaches based on experience.
- Goal-Driven and Theory of Mind Agents demonstrate strategic autonomy, independently determining sub-goals and action sequences to achieve high-level objectives.
- Self-Aware Agents potentially exhibit value autonomy, refining their own objectives and evaluation criteria through reflection and experience.

The progression toward greater autonomy raises important questions about appropriate autonomy levels for different applications, control mechanisms, and value alignment. As noted by Ref. [189], “the problem is not that machines might disobey us but that they might obey us too well” when objectives are misspecified or incomplete. This arises because agentic systems are often designed with predefined goals that lack alignment with human intent. When reward functions fail to capture ethical considerations, agents may overoptimize in unintended ways that violate social norms [190]. Reinforcement Learning from Human Feedback (RLHF) [191] and Inverse Reinforcement Learning (IRL) [192] approaches help align agent behavior with human preferences and demonstrations. Additionally, methods like Constitutional AI [193] embed normative rules into the agent’s decision-making process, while interpretability mechanisms offer external override capabilities when the agent deviates from safe behavior. However, these approaches have limitations for agentic systems. For example, RLHF does not scale well for long-horizon tasks, and human feedback is often noisy or inconsistent. Agents may still exhibit deceptive or manipulative behavior when internal goals conflict with aligned behavior. Distribution shifts between training and deployment further reduce performance, and the agent’s internal reasoning remains largely unobservable and hard to interpret.

Future research is now shifting toward more proactive and intrinsic alignment strategies. One promising direction is Eliciting Latent Knowledge (ELK) [194], which aims to extract truthful internal representations from models rather than relying solely on observable output. Another is the development of self-reflective agents [195] or debate-style reasoning frameworks [196], where agents reason about and critique their own plans before acting, surfacing contradictions or unsafe assumptions. Additionally, researchers are exploring multi-agent oversight networks, where agents with diverse roles and capabilities cross-validate each other’s actions, improving robustness and reducing single-point failure [197]. Embedding norm-tracking memory graphs—explicit, evolving representations of values, constraints, and context may further help agents retain alignment over time and across tasks. Collectively, these directions signal a move toward autonomous, verifiable, and continually aligning agentic AI systems capable of operating safely in complex real-world environments.

3.3.6. Social intelligence

Social capabilities range from non-existent in basic reactive systems to sophisticated social reasoning in advanced agents.

- Reactive and Proactive Agents lack social awareness, operating without models of other agents or humans.
- Limited Memory and Model-Based Agents show basic social adaptation through learned interaction patterns but lack explicit models of others’ mental states.

- Goal-Driven Agents display strategic social behavior, modeling others as elements of the environment that respond to actions according to predictable patterns.
- Theory of Mind and Self-Aware Agents present sophisticated social intelligence, including mental state attribution, perspective-taking, and adaptation to social norms and expectations.

Recent advances in LLM-based agents have shown promising social capabilities, including

1. **Natural language interaction:** Enabling more intuitive human-AI collaboration.
2. **Personality consistency:** Maintaining coherent interaction patterns over time.
3. **Cultural awareness:** Adapting communication to social and cultural contexts.
4. **Emotional intelligence:** Recognizing and responding to human emotional states.

However, challenges remain in developing systems that can navigate complex social dynamics, understand implicit communication, and adapt appropriately to diverse cultural contexts and individual preferences. These challenges stem from the fact that early agentic systems focused primarily on task execution rather than human interaction. They reacted to inputs without understanding people’s thoughts or emotions, and their responses were rigid, lacking personalization and cultural sensitivity. As a result, interactions felt robotic, limited, and socially unaware. As this area is still novel, researchers are actively exploring it by incorporating natural language interaction fluid human-agent collaboration [198] and personality modeling is used to ensure consistent behavior [199]. MetaMind [200] use Theory-of-Mind architectures with specialized agents for mental state inference, cultural norms, and response generation, increasing social reasoning by 35.7%. Culturally adaptive systems such as CLCA [201] improve alignment through simulated role play to internalize implicit norms, while continual personalization through reinforcement learning [202] tailors social behavior based on long-term feedback. Emerging architectural innovations propose ‘Wisdom Layers’ meta-agent systems embedding reflective social cognition and causal ethics into agentic AI [203].

For agentic AI to function culturally intelligently, future systems must embrace adaptability, ethics, emotional awareness, and continual learning. For instance, use of layered agent designs such as Wisdom Layers combined with MetaMind’s separation of Theory-of-Mind reasoning and action generation to improve coordination. In addition, integrate metacognitive modules that handle ethics, uncertainty, and long-term consequences. Additionally, researchers can explore virtual societies, meta-learning, and affective-causal reasoning to develop advanced social strategies and emotionally aware behavior of agentic AI systems.

3.4. Environments of agentic AI

The environments of agentic AI refer to the external or operational settings in which agentic AI systems operate. These examine the nature of the environment and their impact on AI behavior. Structurally, world environments are inherently complex and require agents to balance adaptability. For example, autonomous vehicles must navigate unpredictable traffic patterns [204], yet over adaptation risks instability in different scenarios. Table 7 presents a detailed overview of the various types of environment and their influence on AI behavior. Agentic AI is divided into two distinct phases: Physical agents [205] and virtual agents [206]. Physical environments, which are real-world autonomous systems like robots and self-driving cars, require real-time adaptability, safety, and interaction with dynamic, unstructured environments. While, virtual environments involve simulated tasks like game-playing AI or virtual assistants, offering scalable, controlled platforms for training and testing. Additionally, Table 8 compares key aspects of agentic AI.

Table 7
Environment types in agentic AI.

Environment type	Description	Key considerations	Impact on AI behavior
Static [207]	Environment remains constant or changes minimally over time, allowing stable assumptions about its state transitions.	• Relatively straightforward predictive modeling. • Simpler to plan and reason about. • May lead to brittleness if unforeseen changes occur.	• Agents can rely on precomputed solutions. • Limited adaptability in real-world dynamic scenarios.
Dynamic [207]	Environment evolves over time, often unpredictably, requiring continuous adaptation.	• Incremental or online learning may be necessary. • Real-time monitoring of changes is crucial. • Adaptive mechanisms must handle uncertainty.	• Agents must update models and policies on-the-fly. • Higher computational demands to maintain situational awareness.
Fully observable [208]	Agent has complete information about the environment at every decision point.	• No hidden variables or uncertainty in observations. • Large state spaces can still pose computational challenges.	• Easier decision-making due to full knowledge. • Simplifies policy derivation.
Partially observable [209]	Certain aspects of the environment remain hidden, requiring inference to make decisions.	• Necessitates belief-state tracking or probabilistic models. • Sensor quality and placement become critical.	• Agents face uncertainty about the true state. • Decision-making must incorporate inference, increasing model complexity.
Deterministic [210]	Every action leads to a predictable outcome, with no randomness in transitions.	• Straightforward planning under defined causal structures. • May overlook real-world stochastic variations.	• Classical planning methods are often sufficient. • Lack of robustness to unexpected variability.
Stochastic [211]	Environmental outcomes are probabilistic, so identical actions may lead to different results.	• Must model uncertainty and outcome distributions. • Strategies often center on maximizing expected utility.	• Agents account for randomness. • Policies involve risk management and probabilistic reasoning.
Single-agent [212]	A single agent operates without direct interaction with other AI or human agents.	• Focus on optimizing individual performance. • No need to model other agents' intentions.	• Straightforward decision processes. • Limited realism in multi-entity contexts.
Multi-agent [213]	Multiple agents share the environment, with cooperative, competitive, or mixed objectives.	• Necessitates interaction protocols and strategies. • Can involve negotiation, coordination, or adversarial tactics.	• Agents must account for others' goals and actions. • Emergent dynamics arise from cooperation or competition.

Table 8
Comparison of patterns, types, and environments of agentic AI: Key aspects.

Aspect	Patterns	Types	Environments
Description	Design frameworks and behaviors	Classifications of AI agents	Operational settings
Focus	How agents are designed?	What kinds of agents exist?	Where agents operate?
Examples	Reflection, Tool Use Patterns	Reactive, Learning, Hybrid	Static, Dynamic, Stochastic
Purpose	Structural methodologies	Functional distinctions	Environmental impact

4. Discussion

This section examines the analysis, key challenges, and future directions in agentic AI research, focusing on evaluation difficulties, reliability issues, and problems inherited from the language models that power these agents.

4.1. Analysis

Agentic AI is defined as autonomous systems capable of perceiving their environment, reasoning, and taking actions to achieve specific goals. These systems exhibit a level of agency akin to human decision-making, enabling them to operate independently without constant human intervention. The operational framework of agentic AI integrates core intelligence, back-end decision making, and frontend interaction, allowing the system to continuously refine its operations through feedback loops. This framework facilitates adaptive and intelligent decision making, distinguishing agentic AI from traditional AI systems that rely on explicit programming and human oversight.

The evolution of agentic AI is marked by distinct phases, reflecting advancements in artificial intelligence, machine learning, and cognitive architectures. The early foundations (1980s–1990s) were rooted

in symbolic reasoning and behavior-based AI, with a focus on interaction with the environment rather than complex internal models. Key milestones during this period included the development of rational agents, multiagent systems (MAS), and reinforcement learning (RL), which laid the groundwork for autonomous decision-making. The rise of learning-based agents (2001–2010) saw the integration of machine learning techniques, particularly neural networks, and reinforcement learning, enabling more adaptive and scalable systems. This era also witnessed advancements in real-time object detection, multiagent reinforcement learning (MARL), and unsupervised learning, which further enhanced the capabilities of agentic AI. The deep reinforcement learning era (2011–2017) marked a significant turning point, with the advent of deep reinforcement learning (DRL) enabling systems like deep Q-networks (DQN) to achieve human-level performance in complex tasks, such as playing video games and mastering strategic games like Go. This period also saw the emergence of self-improving agents capable of learning and adapting through self-play. The emergence of general-purpose agents (2018–2021) was characterized by the introduction of large language models (LLMs) like GPT-3, which demonstrated the potential of general-purpose agents capable of reasoning and planning in diverse domains. This era marked a shift toward goal-directed capabilities and emergent cooperation among AI

agents. In the current era (2022-present), the focus has shifted to multi modal agents, human-AI collaboration, and ethical considerations, with models like GPT-4 and HuggingGPT pushing the boundaries of agentic AI by enabling more autonomous, adaptable, and context-aware systems. The paper identifies four key patterns of agentic AI—tool use, reflection, planning, and multi-agent collaboration (MAC) which describe how these systems interact, process tasks, and respond to user inputs. Additionally, agentic AI systems are categorized into seven types based on their capabilities and operational styles: reactive agents, proactive agents, limited-memory agents, model-based agents, goal-driven agents, theory of mind (ToM) agents, and self-aware agents. Each type represents a distinct approach to autonomy and decision-making, with theory of mind (ToM) agents and self-aware agents standing out for their ability to infer human emotions, beliefs, and intentions, and to reason about their internal states, respectively. These capabilities highlight the ethical and social dimensions of agentic AI, as these systems are designed to interact with humans in more natural and empathetic ways. Tables 9 and 10 show a deeper analysis of agentic AI, emphasizing critical insights and reinforcing the study’s analytical framework and contextual understanding.

4.2. Challenges

The analysis also brings attention to subtle, yet crucial, challenges that may hinder progress in the field. The following outline these challenges in detail.

- 1. The evolving goals of agentic AI must align with diverse human values, but human intentions are unpredictable, resulting in unintended outcomes. Multifaceted, context-dependent goals, such as in medical ethics, complicate alignment with broader ethical

principles. Solutions such as value alignment and inverse reinforcement learning are promising but remain underdeveloped for dynamic human objectives.

- 2. Agentic AI must adapt to unpredictable environments like financial markets or autonomous driving, often with incomplete data. Techniques like meta-learning and reinforcement learning improve adaptability but lack consistent robustness. Real-time applications face challenges due to high computational demands and unpredictable scenarios.
- 3. Agentic AI requires significant computational power, energy, and advanced hardware for training and operation, raising costs. Scaling increases demands on data storage, processing, and bandwidth, especially in centralized systems. Innovations in hardware optimization and energy-efficient algorithms are critical, but are still under development.
- 4. Scaling agentic AI in complex systems like smart cities or healthcare involves managing massive datasets and multiple agents without performance degradation. Challenges include goal conflicts, resource demands, and adaptability between subsystems. Decentralized control and federated learning are potential solutions, but face technical barriers to large-scale implementation.
- 5. Agentic AI relies on sensitive data, raising privacy concerns and risks of misuse or breaches. Cybersecurity measures such as differential privacy are essential but must balance functionality with robust protection.
- 6. Agent evaluation lacks standardized benchmarks, as teams often create complex and custom tests scored manually. This approach ensures quality, but introduces bias and struggles with consistency due to model variability.
- 7. Data contamination in LLM training skews benchmark results, worsened by static datasets that fail to match evolving model

Table 9
Comprehensive analysis of agentic AI.

Category	Subcategory	Details
SWOT analysis	Strengths	- Enables independent perception, reasoning, and action, achieving significant efficiency gains. - Leverages reinforcement learning (RL) and large language models (LLMs) for context-aware adjustments, excelling in dynamic, multi-variable tasks. - Multi-agent systems enhance collective problem-solving, demonstrated in coordinated defense strategies and distributed robotics applications.
	Weaknesses	- High computational overhead, particularly in model-based agents, restricts scalability and real-time deployment. - Autonomous decision-making yields inconsistent outcomes in critical, high-stakes scenarios, such as medical emergencies or financial forecasting.
	Opportunities	- Performance hinges on access to high-quality, voluminous datasets. - Offers profound potential to revolutionize sectors, including autonomous surgical systems in healthcare, adaptive learning platforms in education, and precision agriculture for sustainable farming etc. - Collaborative frameworks amplify productivity and innovation, as seen in manufacturing co-bots and scientific research assistants.
	Threats	- Potential to shape global standards for autonomous systems. - Risks of misalignment with human values threaten societal harm. - Autonomous data processing exposes sensitive information to breaches. - Widespread automation of skilled roles risks significant job displacement, exacerbating inequality and social unrest. - High costs may exclude SMEs and developing nations. - The rapid advancement of AI contrasts sharply with the slow pace of policy and regulatory adaptation.
Risk assessment	Technical risks	- Intensive resource demands limit deployment to small-scale or controlled environments, impeding broader adoption.
	Ethical risks	- Self-correcting mechanisms, such as Reflexion-style loops, may amplify initial inaccuracies, compromising system integrity.
	Societal risks	- Independent data handling increases exposure to unauthorized access, threatening user confidentiality. - Reliance on flawed or unrepresentative training data perpetuates and exacerbates societal inequities. - Absence of clear legal frameworks for autonomous actions complicates liability attribution. - Disparities in deployment concentrate benefits in affluent regions, widening socioeconomic divides.
Gap analysis	Knowledge gaps	- Limited understanding of how autonomy functions in unstructured, real-world contexts. - Absence of standardized methodologies to ensure alignment with human values across agent types. - Lack of universally accepted benchmarks to evaluate agent efficacy, hindering comparative analysis. - Need for better frameworks to limit anthropomorphism and false emotional attribution.
	Research gaps	- Scarcity of studies on sustained performance and adaptation over extended periods.
	Implementation gaps	- Limited progress in scaling ToM capabilities for complex, real-world human-AI collaboration. - Few deployments outside controlled laboratory settings, limiting evidence of practical utility. - No established protocols for seamless integration across diverse agent types in multi-agent systems. - Undefined policies fail to address the unique challenges of autonomous systems in critical sectors specially in defense, healthcare.

Table 10
Qualitative analysis of agentic AI.

Category	Subcategory	Details
Qualitative analysis	Thematic insights	- The autonomy of agentic AI obscures distinctions between tool and actor, prompting philosophical inquiries into intent, moral agency, and accountability. - The taxonomy of agentic AI (reactive, proactive, limited memory, model-based, goal-driven, theory of mind, self-aware) reveals a progressive enhancement in autonomy, memory, planning, social understanding, and self-reflection, demonstrating a structured evolution toward higher intelligence. - Autonomous systems redistribute control from humans to technology, as seen in multi-agent coordination, raising questions of oversight and dependency.
	Stakeholder perceptions	- Researchers regard agentic AI as a pioneering domain, balancing enthusiasm for its potential with wariness of unpredictability and ethical dilemmas. - Industry leaders perceive it as a competitive advantage for automation and efficiency, yet remain cautious of reliability and regulatory uncertainties. - General public exhibit ambivalence, blending fascination with capabilities with apprehension over privacy loss and diminished human agency.
	Sociocultural implications	- Autonomy tests societal trust, necessitating narratives that position agentic AI as a collaborative partner rather than an existential threat. - Ignites debates on AI's societal role, demanding culturally attuned frameworks to reconcile diverse value systems and priorities. - Societal adaptation requires transformative shifts in education, workforce skills, and cultural mindsets to harmonize with increasingly autonomous technologies.
	Narrative Dimensions	- Utopian visions envision agentic AI as a catalyst for human flourishing, solving intractable problems like climate change or disease through autonomous ingenuity. - Dystopian Fears fuel narratives of loss—control, privacy, or humanity where unchecked autonomy leads to surveillance states or existential threats. - Pragmatic advocates a middle path, where agentic AI's potential is harnessed through ethical stewardship and participatory governance, shaping a co-evolutionary future.
Future directions	Technical advancements	- Optimize resource use through hybrid symbolic-neural architectures, reducing computational burdens. - Enhance metacognitive functions to improve error detection and correction, bolstering reliability.
	Research priorities	- Design models for human-AI synergy, enhancing trust and efficacy in different real world scenarios. - Formulate testable, domain-specific alignment models to embed human values into agent behavior.
	Practical implications	- Advance toward AGI by pursuing agents with cross-domain reasoning and creative problem-solving capacities. - Establish comprehensive standards for accountability, transparency, and safety tailored to autonomous AI. - Foster education and dialogue to build trust and ensure equitable access, mitigating societal resistance.

capabilities. Dynamic and synthetic benchmarks aim to counter this but reduce human oversight risks accuracy.

- Many LLM benchmarks test single-step reasoning, missing agents' multi-step and tool-use abilities. Agent-specific tests such as AgentBench assess generalization, but nuanced metrics such as planning robustness remain hard to quantify.
- Benchmarks often focus on logic puzzles or games, leaving their relevance to noisy and diverse real-world tasks unclear. Wild-Bench uses real conversations for breadth, while task-specific ones like SWE-bench limit wider applicability.
- Agents inherit LLM biases, showing harmful tendencies and stealthier outputs despite instructions to diversify perspectives. Evaluating and mitigating these biases requires human oversight and a challenging scalable benchmark design.

4.3. Future

The future of this research focuses on potential advancements and emerging opportunities that can drive further progress in this crucial field. The following are the key areas for future exploration:

- The real world is different and sometimes presents complex scenarios. Intelligent systems are still unable to adapt to unforeseen circumstances. More general and adaptive agentic AI systems can unlock new levels of efficiency by continuously improving and transferring knowledge. Yet agentic systems cannot learn continuously or adapt over time. Robustness and generalizability in agentic AI face several challenges, such as preventing new learning from erasing previously acquired knowledge, data scarcity, distribution shift, and lack of causal understanding. Possible future paradigms include developing models that allow agentic AI systems to acquire new knowledge without forgetting the old like “elastic weight consolidation” and “experience replay” etc. Further, integrating meta-learning with agentic systems to enable rapid adaptation to new domains with minimal data.
- Agentic AI provides human-like interaction and is integrated into human-centric environments. Mostly, its technical performance is evaluated without considering sentiments, ethics, values, and social context. However, there are issues with translating human values into computable goals, accurately understanding human emotions, and adapting to cultural contexts. The promising future for researchers lies in this domain by using value alignment approaches from human feedback to guide AI behavior ethically and socially acceptably. In addition, explainable AI (XAI) and social AI, when incorporated with agentic AI, can help these systems provide transparent decision rationales, respond to emotions, and learn social norms.
- Multi-agent systems have been extensively studied by researchers, but within agentic AI the effective coordination, large-scale management, and integration of multiple agents still present significant challenges that need further exploration. Possible solutions include developing scalable decentralized coordination algorithms for efficient self-organization among agents, creating advanced orchestration platforms like JADE for seamless deployment and management of large agent networks, and establishing open standards and APIs for interoperability and standardized communication across diverse agentic systems.
- Agentic AI has the ability to interpret nuanced real-world information, especially language and context, in real-time. But issues arise in achieving true semantic understanding, predicting events with complete precision, and balancing proactive actions with the risk of false positives. Researchers can have a good opportunity to evaluate this enhanced real-time perception and proactive decision-making agentic AI issue. Researchers can present possible solutions by providing improved reasoning through integration with knowledge graphs, optimized hardware for real-time processing, advancing predictive analytics and causal inference, and combining reinforcement learning with planning to develop optimal proactive approaches.

5. Agentic AI depends heavily on its accessibility, and only its experts are using its platform efficiently. Its practical use will require user-friendly platforms that make agentic AI accessible to non-technical users, enabling broader adoption across industries and organizations of all sizes. But agentic AI architectures are complex in nature and still hard for non-technical and business users to use easily. One of the possible solutions is to build systems that have graphical user interfaces (GUIs) where agents work with minimal or no human interaction. Another can be built-in templates that users can just deploy and use. One possible solution can be creating no-code development platforms and providing comprehensive documentation and training resources along with them.
6. Reactive systems can only respond when an event has occurred. Agentic AI is in its initial stage to react to events proactively, so it can foresee potential issues, predict the future state, and take preventive actions. This will be helpful in many real-time scenarios like risk management, cybersecurity, predictive maintenance, etc. This limitation can open the doors for new research to shift from reactive to proactive agentic AI systems. Researchers can integrate different models to understand cause-and-effect relationships for more anticipatory actions. Further, the combination of predictive models with control theory and optimization techniques can enable agents to execute optimal proactive approaches in different real-time scenarios.
7. The workplace is transforming into an agentic AI-centric environment that helps employees achieve higher productivity and creativity, but there is employee resistance and a lack of trust in the decision-making capabilities of agentic AI. There are also risks related to data security that prevent organizations from deploying sensitive data on agentic AI platforms. Therefore, in the future, subject matter experts (SMEs) in agentic AI can tune organizational processes. Organizations can encourage to leadership role model AI adoption, establish clear business rationales for AI capabilities, and foster a culture that embraces hybrid workforces.
8. For collaborative intelligence, federated and decentralized learning frameworks are needed. These frameworks have heterogeneous data distributions that impact model generalizability, high communication overhead and risks of privacy and security. Additionally, when frameworks manage to coordinate among numerous decentralized nodes scalability remains a concern. Current solutions focus on personalized federated learning to adapt global models to local data characteristics, compression techniques to minimize communication costs, and secure aggregation protocols using advanced cryptographic methods to preserve privacy. But, they still face limitation in solving the adversarial conditions maintaining performance with non-IID data. Therefore, future research can make scalable secure and incentive-compatible frameworks integrated with advanced agentic AI systems. These agentic systems can autonomously manage participant selection based on data quality and reliability, dynamically adjust learning rates and aggregation strategies to optimize convergence, and detect as well as mitigate adversarial behaviors in real-time.

5. Conclusion

This paper provides a comprehensive overview of the evolution, key aspects, and challenges of agentic AI, which refers to autonomous AI systems capable of reasoning, decision-making, and adapting to dynamic environments. The evolution of agentic AI traces back to the 1980s, with foundational work in robotics and cognitive science, and has progressed through the rise of learning-based agents, the deep reinforcement learning era, and the emergence of general-purpose agents like GPT-3. Today, agentic AI is characterized by multi-modal

and collaborative systems that integrate vision, language, and action for more autonomous and context-aware decision-making. The field has experienced a significant acceleration since 2018, with the emergence of large language model-based agents demonstrating capabilities that blur traditional boundaries between narrow and general AI systems. This acceleration shows no signs of slowing, with multi-modal and collaborative agent architectures representing the current frontier of development.

Our taxonomic framework identifies a progressive spectrum of agent types, from reactive to self-aware systems, with each type building upon and extending the capabilities of previous generations. This taxonomy provides a structured approach for understanding the diverse landscape of agentic systems and identifying capability gaps and development pathways. The paper identifies key patterns of agentic AI, such as tool use, reflection, planning, and multi-agent collaboration, and classifies agents into types like reactive, proactive, and self-aware agents, each suited for different tasks and environments.

Agentic AI is driving transformation by enabling autonomous decision making, personalized services, and optimized resource management. However, significant challenges remain, such as aligning AI goals with human values, ensuring adaptability in unpredictable environments, addressing resource constraints, and managing privacy and security risks. The paper also highlights the need for new regulatory frameworks to address accountability and ethical concerns. Future research will focus on developing multi-modal and collaborative agents, improving scalability and efficiency, and advancing general-purpose agents capable of reasoning and self-improvement. Overall, agentic AI represents a significant leap toward autonomous, context-aware systems, but its safe and effective deployment will require addressing technical, ethical, and regulatory challenges. The paper concludes that agentic AI is steadily moving from passive models to fully autonomous systems capable of reasoning, planning, and adapting in complex environments. As research continues to refine memory mechanisms, alignment strategies, and multi-modal integration, agentic AI is expected to push further into domains requiring human-like creativity, judgment, and autonomy. Although substantial progress has been made, important challenges remain in developing agentic systems that reason reliably, operate robustly across contexts, align with human values, and integrate effectively into existing sociotechnical systems. These challenges span technical, ethical, and governance dimensions, requiring interdisciplinary approaches to address effectively.

CRedit authorship contribution statement

Ume Nisa: Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Muhammad Shirazi:** Writing – review & editing, Formal analysis, Conceptualization. **Mohamed Ali Saip:** Supervision, Methodology. **Muhammad Syafiq Mohd Pozi:** Supervision, Methodology.

Ethical statement

We confirm that this paper presents original research and has not been published previously. Additionally, it is not under review or consideration for publication elsewhere. We provide our consent for its submission to this journal.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The author Muhammad Shirazi is currently employed by MEDGULF.

The other authors declare that there are no financial or personal relationships that could be perceived to have influenced the research findings and conclusions presented in this study. Ume Nisa, Mohamed Ali Saip, and Muhammad Syafiq Mohd Pozi are affiliated with School of Computing, Universiti Utara Malaysia.

Data availability

Data sharing is not applicable to this study, as no data was generated during the study.

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